

Table 5. Main planetary parameters derived from the transit and joint RV–TTV analyses

Parameter	L 98-59 b	L 98-59 c	L 98-59 d	L 98-59 e	L 98-59 f
<i>Orbital parameters</i>					
P (days)	$2.2531140 \pm 4 \cdot 10^{-7}$	$3.6906764 \pm 4 \cdot 10^{-7}$	$7.450729 \pm 2 \cdot 10^{-6}$	12.8278 ± 0.0018	23.064 ± 0.055
t_0 (TBJD*)	1366.17056 ± 0.00024	1367.27303 ± 0.00015	1362.74002 ± 0.00025	1438.48 ± 0.15	1437.84 ± 0.75
a (au)	0.0223 ± 0.0007	0.0309 ± 0.0010	0.0494 ± 0.0016	0.0712 ± 0.0022	0.1052 ± 0.0033
i (°)	$88.08^{+0.23}_{-0.20}$	$88.88^{+0.21}_{-0.17}$	$88.44^{+0.05}_{-0.05}$	$> 80^\dagger$	$> 74^\dagger$
e	$0.031^{+0.017}_{-0.016}$	$0.002^{+0.002}_{-0.001}$	$0.006^{+0.007}_{-0.004}$	$0.012^{+0.009}_{-0.008}$	$0.044^{+0.027}_{-0.028}$
<i>Transit parameters</i>					
b	$0.51^{+0.04}_{-0.05}$	$0.41^{+0.05}_{-0.07}$	$0.921^{+0.007}_{-0.007}$	—	—
δ (ppm)	589 ± 14	1486 ± 30	1445 ± 47	—	—
t_{14} (hours)	1.01 ± 0.03	1.28 ± 0.04	0.84 ± 0.04	—	—
<i>Physical parameters</i>					
R_p ($R_\oplus$)	0.837 ± 0.019	1.329 ± 0.029	1.627 ± 0.041	$1.42^{+0.38 \times}_{-0.08}$	$1.43^{+0.38 \times}_{-0.10}$
M_p ($M_\oplus$)	0.46 ± 0.11	2.00 ± 0.13	1.64 ± 0.07	$2.82 \pm 0.19^\square$	$2.80 \pm 0.30^\square$
ρ ($\text{g} \cdot \text{cm}^{-3}$)	$4.3^{+1.1}_{-1.0}$	$4.7^{+0.5}_{-0.4}$	$2.2^{+0.2}_{-0.2}$	$5.2^{+1.3}_{-2.6}$	$5.0^{+1.7}_{-2.4}$
S ($S_\oplus$)	24.5 ± 2.5	12.7 ± 1.3	4.97 ± 0.52	2.40 ± 0.25	1.10 ± 0.11
$T_{\text{eq}, A_B=0}$ (K)	620 ± 13	526 ± 11	416 ± 9	347 ± 7	285 ± 6

NOTE—*TBJD = BJD − 2457000. $^\dagger 2\sigma$ lower limit with a maximum inclination for planet e of 88.82° and for planet f of 89.20° constrained by the absence of transit. $^\times$ Radius prediction from empirical mass–radius relation (Parviainen et al. 2024).

$^\square$ Minimum mass $M_p \sin i$.